

Anthony Ibarra
Embedded Systems Engineer

(Chicago, IL; Available to Start Onsite)

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Education:

University of Illinois Chicago (UIC)

Jan 2023 - May 2025

Master of Science in Electrical and Computer Engineering Chicago, United States

Relevant Coursework:

- Mechatronics Embedded Design, Adaptive Digital Filters, Linear Systems Theory & Design, Convex Optimization
- Electromechanical Energy Conversion, Audio & Acoustic Signal Processing, Filter Synthesis Design
- Electromagnetic Compatibility, Neural Networks, Advanced Computer Communication Networks

University of Illinois Chicago (UIC)

Aug 2017 - May 2022

Bachelor of Science in Computer Engineering Chicago, United States

Relevant Coursework:

- Embedded Systems, Control Engineering, Principles of Modern Control, Principles of Automatic Control
- Computer Organization, Computer Architecture
- Robotics Algorithm and Control, Pattern Recognition, Computer Comm Networks, Artificial Intelligence

Technical Skills:

Embedded Programming & Firmware:

- Bare-metal development, RTOS (QXK, FreeRTOS), Embedded C/C++, State Machines
- PID Control, Bootloader Implementation, U-Boot, Device Driver Development

Microcontrollers & SoCs:

- Texas Instruments Tiva C, Freescale MCUs, Raspberry Pi, STM32, Espressif, Arduino Nano
- ARM Cortex-M series, QEMU

Protocols & Communication Interfaces:

- UART, SPI, I2C, BLE, TCP/IP, UDP

Testing & Debugging:

- Unit Testing, Test-Driven Development (TDD), Pytest, uTest
- Hardware-in-the-Loop (HIL), Tracing, GDB Debugging
- Oscilloscope, Multimeter, RF Analyzer (FieldFox)

Robotics & Control Systems:

- ROS2, Embedded Vision (OpenCV), Sensor Integration
- Autonomous Control, Control Systems

Tools & IDEs:

- Code Composer Studio, Keil µVision, STM32 CubeMX, Arduino IDE
- PlatformIO, VSCode, MATLAB, Simulink, Quartus, Icarus Verilog, Wireshark
- Git, GitHub, Docker, Ubuntu, MySQL

Programming Languages & Hardware Description:

- C, C++, Python, Bash, MIPS Assembly, ARM Assembly

Projects:

Board Bring-up From the ground up – Embedded Systems

March 2026 - March 2026

(Bare Metal Programming for ARM) | QEMU ARMv7-A & Cortex-A9 MPCore

- Developed bare-metal firmware for ARMv7-A (Cortex-A9 MPCore) using QEMU Versatile Express, simulating board bring-up from reset through application execution.
- Integrated U-Boot bootloader to initialize the system and load firmware images, implementing CMake-based build automation and Bash scripts for compilation, flashing, and debugging workflows.
- Implemented a PL011 UART device driver for the Versatile Express platform, enabling serial communication and runtime debugging.
- Designed and configured interrupt handling using the ARM Generic Interrupt Controller (GIC) on the Cortex-A9 MPCore for interrupt-driven peripheral interaction.
- Built a cooperative task scheduler supporting non-preemptive multitasking for embedded firmware tasks.
- Performed remote debugging with GDB through WSL Bash terminal, connecting to QEMU for step execution, register inspection, and memory analysis.

Bootloader & Firmware Updater – Embedded Systems
(Blinky to Bootloader: Bare Metal Programming YT)

Nov 2025 - Nov 2025

- Developed a secure bare-metal bootloader and firmware update system for a microcontroller-based platform.
- Implemented AES-based CBC-MAC firmware authentication to verify firmware integrity and prevent unauthorized updates.
- Designed a bootloader state machine with timeout handling to manage firmware update flow, error recovery, and system initialization.
- Implemented flash memory management routines to safely erase and program application firmware during updates.
- Developed a full UART communication driver with interrupt-safe ring buffer implementation to prevent race conditions during packet transmission and reception.
- Designed a custom UART packet protocol including ACK/NACK responses and CRC-based error detection for reliable firmware transfer.

Autonomous-Driving Car – Mechatronics Embedded Design
University of Illinois Chicago (UIC) – Chicago, IL

Jan 2023 - May 2023

- Led a multidisciplinary engineering team in the design and development of an autonomous embedded system, coordinating hardware, firmware, and testing milestones.
- Designed and implemented a motor driver circuit using MOSFET drivers (single FET, half-bridge, and full H-bridge configurations) controlled via PWM signals from a microcontroller for DC motor actuation.
- Developed a boost converter power supply to support multiple voltage rails for sensors, control electronics, and motor drivers.
- Implemented and tuned PID control algorithms in embedded firmware to regulate steering (servo motor) and vehicle velocity (DC motor) using encoder feedback.
- Integrated line-tracking camera sensors and wheel encoders with the microcontroller, implementing sensor filtering and signal processing in embedded software.
- Performed hardware validation and debugging using oscilloscopes, multimeters, and benchtop power supplies to analyze PWM signals, power electronics behavior, and sensor outputs.

Automated Watering Systems, Senior Design Product Development
University of Illinois Chicago (UIC) Chicago, United States

Aug 2021 - May 2022

- Developed an IoT-based automated plant watering system using an Arduino Nano 33 IoT, integrating soil moisture sensors, plant data, and weather conditions to determine watering decisions.
- Developed C/C++ firmware for Arduino Nano 33 IoT, acquiring soil moisture data and automating irrigation decisions based on thresholds and weather API integration.
- Designed wireless communication architecture enabling device monitoring and control via Bluetooth Low Energy (BLE).
- Built a UDP client-server communication layer between the embedded system and a mobile application for remote monitoring and manual override.
- Developed a cross-platform mobile application using the Kivy framework to display real-time sensor data, weather conditions, and system status.
- Collaborated within a 4-member engineering team, coordinating development tasks and producing weekly technical progress reports.
- Prototyped and tested an integrated embedded hardware-software system combining sensors (moisture sensors), wireless connectivity (Bluetooth), and mobile interface.
- Integrated external weather API data with local soil moisture readings to dynamically adjust watering schedules.

Certifications:

- C Programming for Embedded Applications - LinkedIn Learning
- Introduction to FreeRTOS - LinkedIn Learning
- Learning FPGA Development, Verilog for FPGA, Learning LabVIEW - LinkedIn Learning
- Getting Started with RISC-V - LinkedIn Learning
- Creating and Securing Bluetooth Low Energy (BLE) Application - LinkedIn Learning